MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6303

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Rectifier or shunt diode, high-vacuum

See figure 1

Mounting position: Vertical, base down

Weight: 10-ounces (283.5 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	Ieff	Pp
Unit:	Vac	kv	a	mAdc	Aac	W
Rectifier						
Maximum:	11.8	40	2.5	700	---	550
Minimum:	11.2	---	---	---	---	---
Shunt diode						
Maximum:	12.5	33	50	---	1.25	500
Minimum:	11.9	---	---	---	---	---
TEST CONDITIONS:	11.5	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1301	Filament current	-		0.65	II	If	14.00	16.25	Aac
1256	Electrode current (anode)	-	Eb = 500 Vdc	0.65	II	Ib	750	---	mAdc
1231	Peak emission	-	eb = 8.0 kv; Ef = 12.2 Vac	0.65	II	is	25	---	a
1353	Operation of rectifiers (1)	1	epx = 40 kv; Ib = 700 mAdc; ib = 2.5 a	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
1041	Shock	-	Hammer angle = 20°	---	---	---	---	---	---
1353	Operation of rectifiers (2)	1,2	Ef = 12.2 Vac	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group C; operation of rectifiers (1); t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
1231	Peak emission	-		---	---	is	20	---	a

NOTES:

1. The tube shall operate for 2 minutes without arcing or sign of gas during a test interval not to exceed 5 minutes.
2. The tube shall be operated as a shunt diode in a thyratron modulator. The circuit constants shall be chosen to give the following diode operating conditions: epx = 33 kv; ib = 50 a; and effective current (Ieff) = 1.25 Aac.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

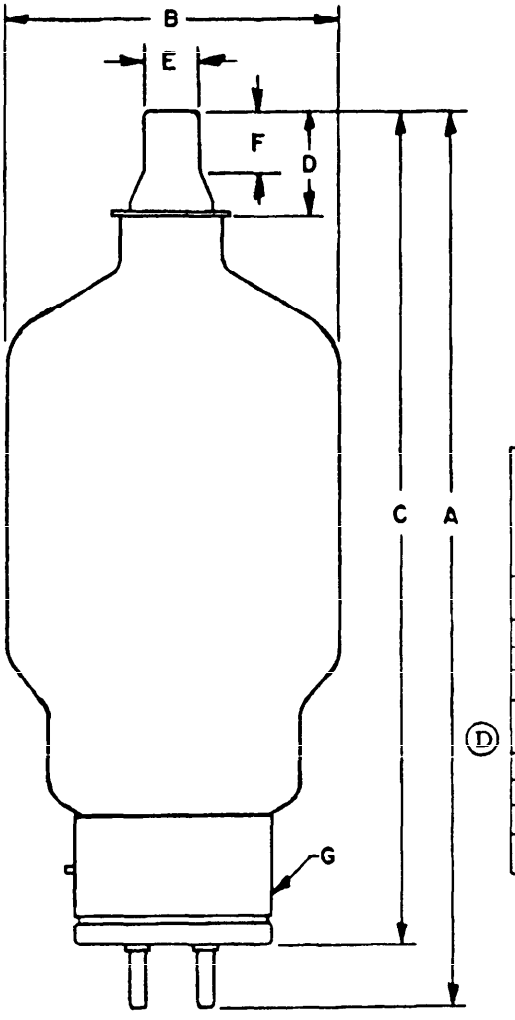
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2952)

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PIN CONNECTIONS	
Pin No.	Element
1	nc
2	f
3	f
4	nc
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	9.250 (234.95)	9.750 (247.65)
B	---	3.625 (92.08)
C	8.472 (215.19)	9.032 (229.41)
Quality conformance inspection, part 3 (see note)		
D	.969 (24.62)	1.031 (26.19)
E	.559 (14.20)	.573 (14.55)
F	.594 (15.09)	.656 (16.66)
G	Base: A4-18 (EIA)	

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 6303.

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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NAVY DEPARTMENT
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NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-26254

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